Xedni affine-lift optimization: seven- and ten-coefficient paths

Date: 2026-11-01 Updated: 2026-11-02

Outcome

The coefficient-lifting seam of prime-field xedni now has two executable, certifying paths exposed as `ca-icx xedni-lift`. Both use high-precision LLL, **exact-rational** Babai closest-plane selection, and a bounded exact search. The seven-coefficient Weierstrass path and the ten-coefficient general-cubic path both rank by an exact discriminant for all `2 <= r <= 9`. The latter uses Fisher's direct ternary-cubic Hessian determinant, so Step 9 no longer depends on first finding a Step-8 Weierstrass conversion. Both paths reject any direction that changes a supplied point-incidence row or coefficient residue class and recheck those properties on the result.

The frozen p=67 comparison has **not run**. The mandatory pre-run checkpoint upload is denied by both the installed presigned token (HTTP 403) and the 2026-11-02 managed AWS identity (`PutObject AccessDenied`). The exact blocker and exit condition are recorded in `problems/xedni-affine-cvp-checkpoint-access.md`. Consequently this report makes no empirical claim that the new nonlinear search beats the 2000 implementation.

Claim labels used below are **source**, **derived**, **verified control**, **implemented but unmeasured**, and **blocked**.

What "the nine points" actually means

Source. JKSST Section 3.2 chooses an integer `r` with `2 <= r <= 9`, saying that `r=4..6` is most likely. A general projective cubic has ten coefficients up to the incidence equations, so nine independent point rows are the maximum the construction permits; the algorithm does not require a special fixed set of nine points.

For `r>=4` in general cubic form, the paper first sends four points to a fixed projective frame. Steps 6–7 lift the remaining point coordinates and solve an affine coefficient system. Steps 9–10 vary kernel vectors to reduce the discriminant and apply a rank heuristic. The authors' implementation already used the Havas–Majewski–Matthews HNF algorithm. Therefore replacing a raw integer solve by "some lattice reduction" is not itself a new improvement over their implementation.

Implemented algorithms

Let `u*` be one feasible coefficient vector, let the rows of `D` be admissible coefficient changes, let `B` be the lifted-point incidence matrix, and let `q=Mp` be the required modulus. The input certificate checks

`B D_i = 0` and `D_i = 0 (mod q)`

for every direction. The optimizer then:

1. LLL-reduces `D` with the repository's 2048-bit fixed-point Gram–Schmidt implementation at delta 0.99;
2. applies exact-rational Gram–Schmidt and Babai nearest-plane to target `-u*`, retaining arbitrary-precision integer center coefficients;
3. enumerates a bounded neighborhood around the Babai coefficients;
4. uses the model-specific exact objective; and
5. certifies `B u = B u*` and `u = u* (mod q)` on the selected vector.

For both models, step 4 rejects singular curves and orders candidates by exact discriminant bit length, coefficient-height bits, then exact squared L2 norm. The general-cubic machine result now emits `discriminant_evaluated: true`.

Source and implementation. Fisher Section 7.2 gives a direct construction for a ternary cubic `U`: compute its integral Hessian `H(U) = -det(d^2 U)/2`; put the six quadratic coefficient rows `U_X,U_Y,U_Z,H_X,H_Y,H_Z` into a `6 x 6` matrix; and take its determinant. This determinant is `+/-`

1728 Delta. The implementation evaluates that matrix with fraction-free Bareiss elimination, so it detects singularity and supplies the Step-9 objective using integers only.

This deliberately bypasses, rather than claims to implement, JKSST Step 8. A projective linear change sends a plane cubic to Weierstrass form only when a rational flex is available; an arbitrary rational point does not supply that condition. A separate birational conversion or direct plane-cubic group law is still required by downstream rank/dependence code, but neither is needed to minimize the model discriminant.

Derived. For signed input order (a0,a1,a3,-a00,-a2,-a4,-a6), completing the square gives

```
(2 a0 y + a1 x + a3)^2 = A x^3 + C2 x^2 + C1 x + C0,
A  = 4 a0 a00,
C2 = 4 a0 a2 + a1^2,
C1 = 4 a0 a4 + 2 a1 a3,
C0 = 4 a0 a6 + a3^2.
```

The implementation evaluates the standard integral cubic discriminant 18 A C2 C1 C0 - 4 C2^3 C0 + C2^2 C1^2 - 4 A C1^3 - 27 A^2 C0^2 exactly. It is 16 times the usual elliptic discriminant in the monic standard case, so it preserves ordering there and also detects singularity in this generalized form when the leading coefficients are nonzero.

The data flow and claim boundary are shown in the labeled editable diagram and its SVG.

Verification completed

Verified control. Under the suite's declared Rust 1.87.0 floor, the 2026-11-02 focused validation currently establishes:

- the focused library and CLI controls for both models pass;
- both actual ca-icx JSON integration paths pass;
- a synthetic rank-three coset containing the known curve y^2+y=x^3-x recovers its signed vector (1,0,1,-1,0,1,0) while preserving four incidence rows and modulus 67;
- invalid incidence-changing and noncongruent directions are rejected before search; and
- the discriminant formula returns 1024 for y^2=x^3-x, exactly 16 times its elliptic discriminant 64;
- the direct ternary formula returns -110592 = -1728 * 64 for the same Weierstrass plane cubic, returns 34,012,224 for the Fermat cubic, and returns zero for the singular Hesse member x^3+y^3+z^3-3xyz;
- exact Babai recovers a representative whose center coefficient has magnitude 10^40, removing the former i64 limit; and
- nine rational multiples on y^2=x^3-x+1 produce an exact rank-nine 9 x 10 incidence matrix. General-cubic mode recovers the coefficient vector (1,0,0,0,0,0,-1,-1,0,1) while preserving all nine rows and modulus 67; its direct determinant is 635,904, equal in magnitude to 1728 * 368 for the planted curve y^2=x^3-x+1.

The complete current-tree release library suite also passes: 3,052 passed, 0 failed, and 90 ignored in 239.37 seconds. Direct touched-file rustfmt checks, cargo clippy --all-targets -- -D warnings, and both release CLI integration paths pass on the same tree.

For orientation only, the synthetic input has a 118-bit exact discriminant and 15-bit maximum coefficient, while the planted short representative has a 10-bit exact discriminant and 1-bit maximum coefficient. This is a planted correctness control, not evidence about natural xedni lift quality.

The nine-point fixture is likewise a planted correctness control. Its scalar input representative has 14-bit maximum coefficient; exact Babai and radius zero recover a 1-bit representative. The editable quantitative source, SVG, and source data nine_point_control.csv show that exact deterministic comparison. It is not a comparison with HMM-HNF and not a natural-lift result.

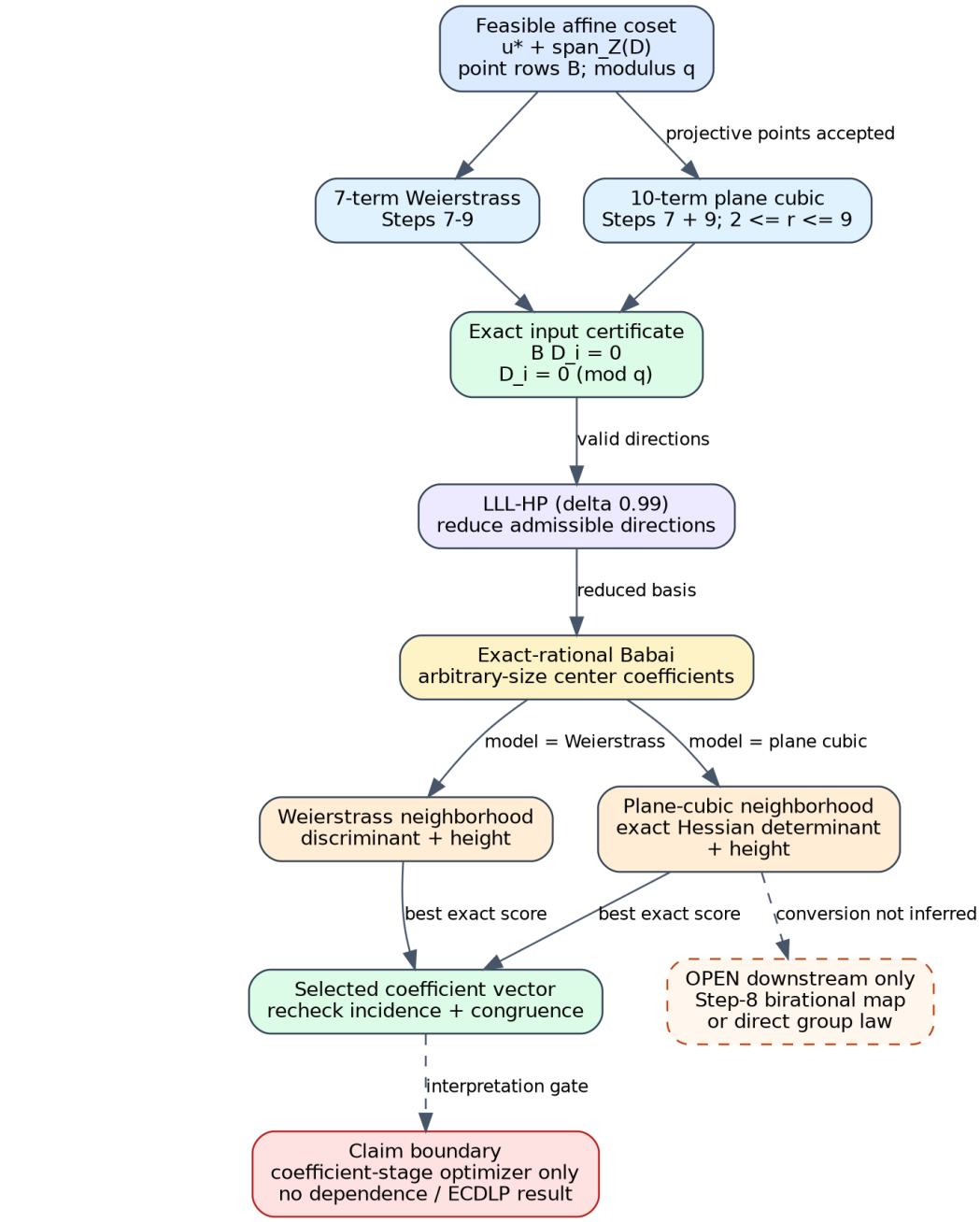


Figure 1: Certified affine-lift pipeline

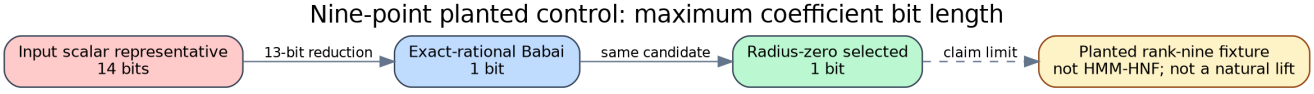


Figure 2: Nine-point planted coefficient-height control

Frozen diagnostic and current evidence boundary

The protocol freezes one JKSST-derived p=67 coefficient coset: curve y^2=x^3+28, generator (1,30), multipliers (1,5,17,29), balanced point lifts, four exact incidence rows, and radii 0,1,2,4. Its three rows of interest are:

Row	Meaning	Status
raw exact solve	deliberately skew Smith-decomposition representative	input frozen; not a fair HMM baseline
LLL + Babai	modern linear coefficient-height control	implemented but unmeasured on the frozen coset
exact neighborhood	nonlinear discriminant candidate	implemented but unmeasured on the frozen coset

Blocked. No quantitative graph is emitted for the p=67 comparison because there are no run rows to graph. Inventing a plot from planned values or silently substituting the planted control would be misleading. Once the checkpoint gate passes, `run_evaluation.py` produces the raw JSON and the quantitative graph will be generated from those retained rows.

Relation to the asymptotic obstruction and other fields

Source. JKSST Theorem 4.1 conditionally bounds the original prime-field xedni success probability by C0/p; their experiments found useful dependent lifts prohibitively rare. Lowering one curve discriminant is therefore a constant-factor engineering result unless a variant explicitly breaks one of the theorem's hypotheses. This optimizer does not claim to do so and does not test point dependence.

Derived. The implementation is prime-field Step-7/9 machinery. It does not construct a lift for F_{2^m}. A global number field whose residue field contains general F_{2^m} coordinates must have degree at least m; Koblitz curve coefficients descending to F_2 do not make their ECDLP point coordinates descend. The unavailable Kogure 2000/2002 primary texts remain a source gate for a source-faithful characteristic-two variant.

Requirement status

Requested item	Status	Evidence / remaining gap
Reconstruct the lift and "nine point" step	verified complete through direct Step-9 discriminant search	JKSST Section 3.2; ten coefficients, direct point rows, rank-nine control, Fisher determinant
Implement a modern lift optimizer	verified complete for both coefficient models	module, exact-rational Babai, two exact discriminant objectives, exact certificates, focused tests
General-cubic Step 8/conversion	bypassed for discriminant; conversion still open	direct Hessian determinant supplies +/-1728 Delta; downstream birational map/group law is separate
Test whether it drastically improves the old algorithm	partial / blocked	frozen p=67 protocol exists; S3 checkpoint gate cannot upload
Prime-field ECDLP improvement	not established	no dependence test or scalar recovery
Binary and Koblitz transfer	not implemented	degree-m lift plus Kogure primary-text gate remain

Sources

1. M. J. Jacobson Jr., N. Koblitz, J. H. Silverman, A. Stein, and E. Teske, “Analysis of the xedni calculus attack,” *Designs, Codes and Cryptography* 20 (2000), 41–64, DOI 10.1023/A:1008312401197. Primary PDF reviewed, especially Sections 3.2, 4, and 5.4.
2. J. H. Silverman, “The xedni calculus and the elliptic curve discrete logarithm problem,” *Designs, Codes and Cryptography* 20 (2000), 5–40, DOI 10.1023/A:1008319518035.
3. G. Havas, B. Majewski, and K. Matthews, “Extended GCD and Hermite normal form algorithms via lattice basis reduction,” *Experimental Mathematics* 7 (1998), 125–136.
4. T. Fisher, “The invariants of a genus one curve,” *Proceedings of the London Mathematical Society* 97 (2008), 753–782, DOI 10.1112/plms/pdn021; arXiv:math/0610318. Primary PDF reviewed, especially Theorem 4.4 and Section 7.2.
5. Jun Kogure, “Implementation of the Xedni Calculus algorithm,” IEICE Technical Report 100(77), 17–23 (2000), NDL 5409112; metadata only.
6. Jun Kogure, “xedni calculus method (characteristic 2),” ISEC2002-10, 57–60 (2002), NDL 6188214; metadata only.

Reproduction and files

- Code: `suite/src/cryptanalysis/xedni_lift.rs`
- CLI: `ca-icx --json xedni-lift --model weierstrass|plane-cubic --input FILE`
- Nine-point fixture: `suite/tests/fixtures/xedni_lift_nine_points.json`
- Frozen input: `inputs/p67_balanced_snf.json`
- Runner: `run_evaluation.py` (not yet executed)
- Blocker: `problems/xedni-affine-cvp-checkpoint-access.md`
- This PDF: `xedni_lift_optimizer_20261101.pdf`